

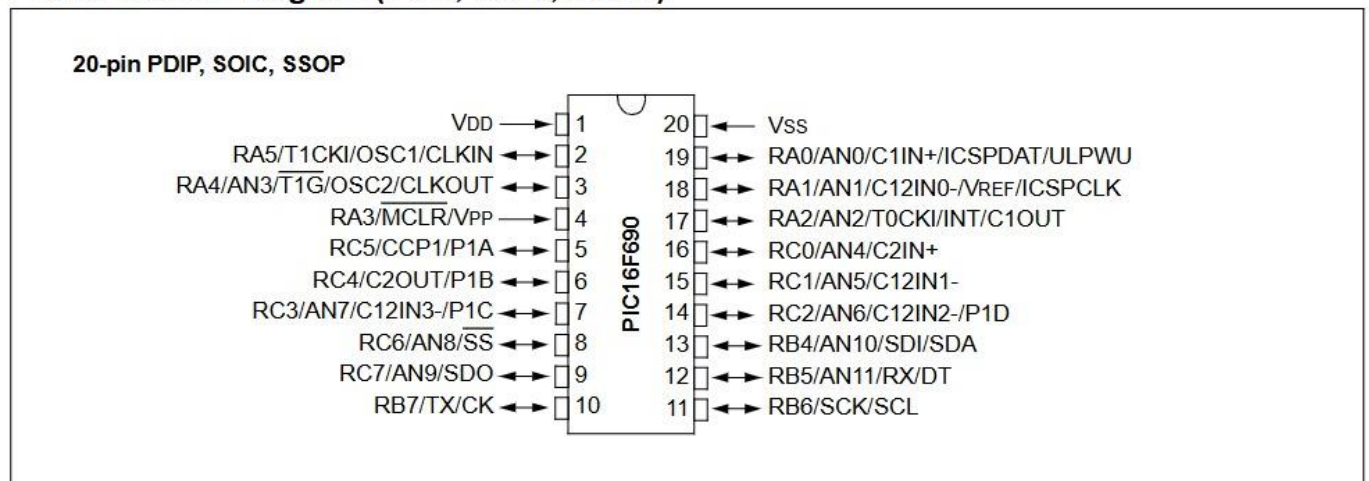
Project Report: LED Blinking using PIC16F690 and PICkit 3

Date: 29.04.26

1. Introduction

This report details the design, hardware configuration, and software implementation of a basic embedded system project: blinking an LED using the PIC16F690 microcontroller. The project utilizes the PICkit 3 In-Circuit Serial Programmer (ICSP) and the MPLAB X V6.20 Integrated Development Environment (IDE) to demonstrate the fundamental principles of microcontroller programming and hardware interfacing.

PIC16F690 Pin Diagram (PDIP, SOIC, SSOP)



The **PIC16F690** is a versatile **20-pin, 8-bit** microcontroller based on the RISC architecture. It operates at a frequency of up to **20 MHz** with an external clock, though it features a highly accurate **8 MHz internal oscillator** that eliminates the need for extra components. The chip includes **7 KB of Flash memory** for code, **256 bytes of RAM**, and **256 bytes of EEPROM** for data storage. It is equipped with **12 analog input channels** (10-bit ADC), dual comparators, and support for multiple communication protocols like **I2C, SPI, and UART**. Operating between **2.0V and 5.5V**, it is well-known for its **nanoWatt technology**, which ensures extremely low power consumption, making it perfect for battery-operated embedded projects.

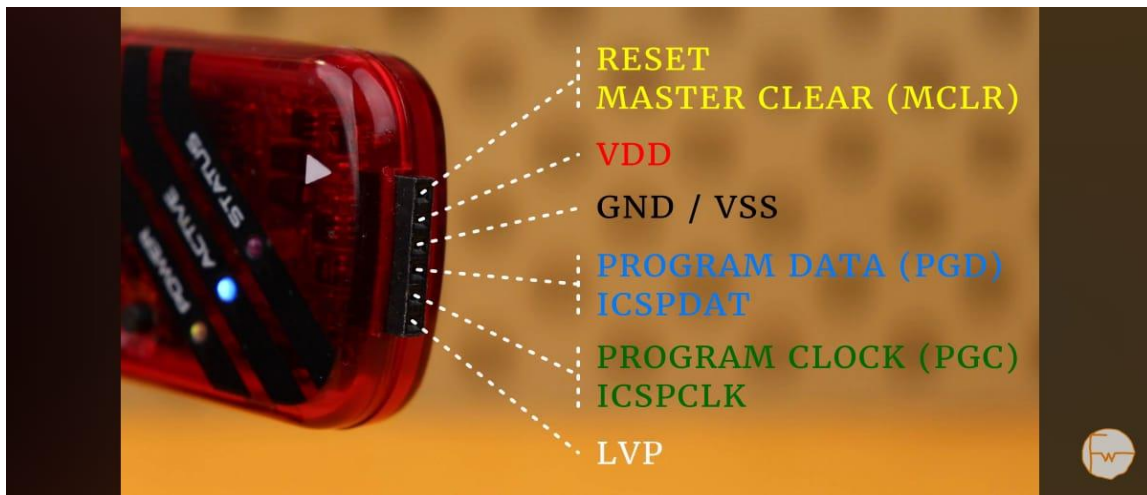
2. Hardware Requirements

The following components are essential for the successful execution of this project:

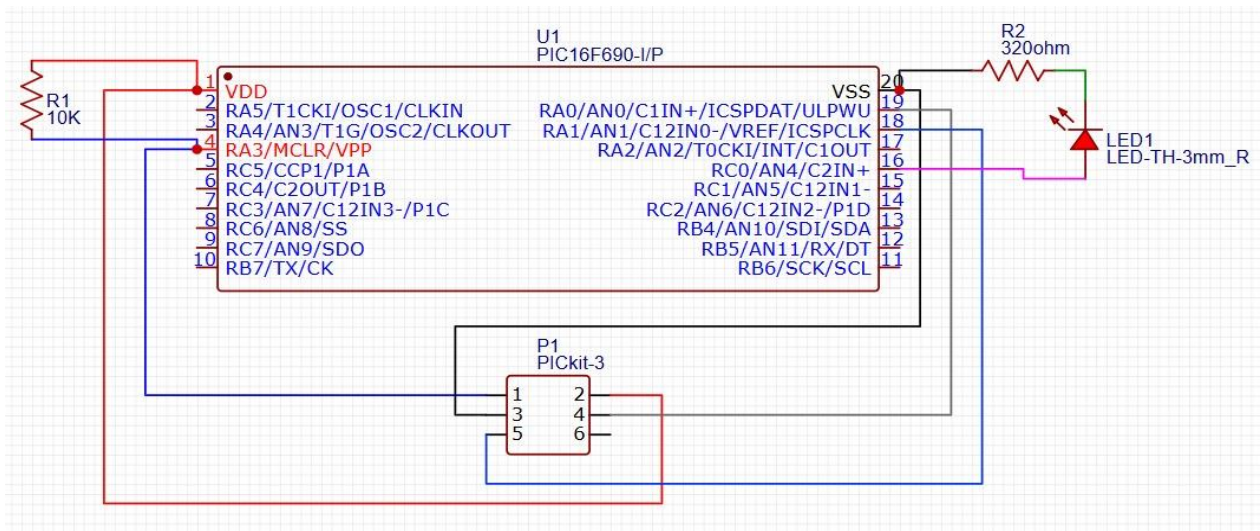
Component	Quantity	Description/Specification
PIC16F690 Microcontroller	1	20-Pin Flash-Based 8-Bit CMOS Microcontroller.
PICkit 3 Programmer	1	Microchip's hardware debugger and programmer.
Breadboard	1	Standard solderless breadboard for circuit prototyping.
LED	1	Standard 5mm LED (e.g., Red or Green).
Resistor (10k Ω)	1	Pull-up resistor for the MCLR (Reset) pin.
Resistor (320 Ω)	1	Current-limiting resistor for the LED.
Jumper Wires	Set 10	Male-to-Male wires for
USB Cable	1	Mini-B USB cable to connect PICkit 3 to PC.

3. Circuit Design

PIN CONFIGURATION OF PICKIT 3



CIRCUIT DIAGRAM OF PIC16F690 WITH PICKIT 3



The project employs In-Circuit Serial Programming (ICSP) to program the chip. The PICKit 3 connects to the PIC16F690 as follows:

- **Pin 1 (Arrow/MCLR):** Connected to Pin 4 of the PIC. A 10kΩ pull-up resistor is connected between Pin 4 and Pin 1 (VDD).
- **Pin 2 (VDD):** Connected to Pin 1 of the PIC (Positive Power).
- **Pin 3 (VSS):** Connected to Pin 20 of the PIC (Ground).
- **Pin 4 (PGD/ICSPDAT):** Connected to Pin 19 of the PIC.
- **Pin 5 (PGC/ICSPCLK):** Connected to Pin 18 of the PIC.

The LED output is configured on **Pin 16 (RC0)**. The long leg (Anode) connects to Pin 16, and the short leg (Cathode) connects to Ground(Pin20) via the 320Ω resistor.

4. Software Used

Using the Software MPLAB X V6.20 The source code is written in C using the XC8 compiler. It includes configuration bits to ensure the internal oscillator is used and safety timers like the Watchdog Timer are disabled.

```
#include <xc.h>

// CONFIGURATION BITS
#pragma config FOSC = INTRCIO    // Internal oscillator
#pragma config WDTE = OFF        // Watchdog Timer Off
#pragma config PWRTE = OFF       // Power-up Timer Off
#pragma config MCLRE = ON        // MCLR Pin On
#pragma config CP = OFF          // Code Protection Off
#pragma config BOREN = ON        // Brown-out Reset On

#define _XTAL_FREQ 4000000      // 4MHz Clock

void main(void) {
    OSCCONbits.IRCF = 0b110;    // Set internal clock to 4MHz
    ANSELbits.ANS4 = 0;         // Disable analog functionality on
RC0
    TRISCbits.TRISC0 = 0;        // Set RC0 (Pin 16) as output
    PORTCbits.RC0 = 0;          // Initialize LED as off

    while(1) {
        PORTCbits.RC0 = 1;      // LED ON
        __delay_ms(5000);       // 5sec delay
        PORTCbits.RC0 = 0;      // LED OFF
        __delay_ms(5000);       // 5sec delay
    }
}
```

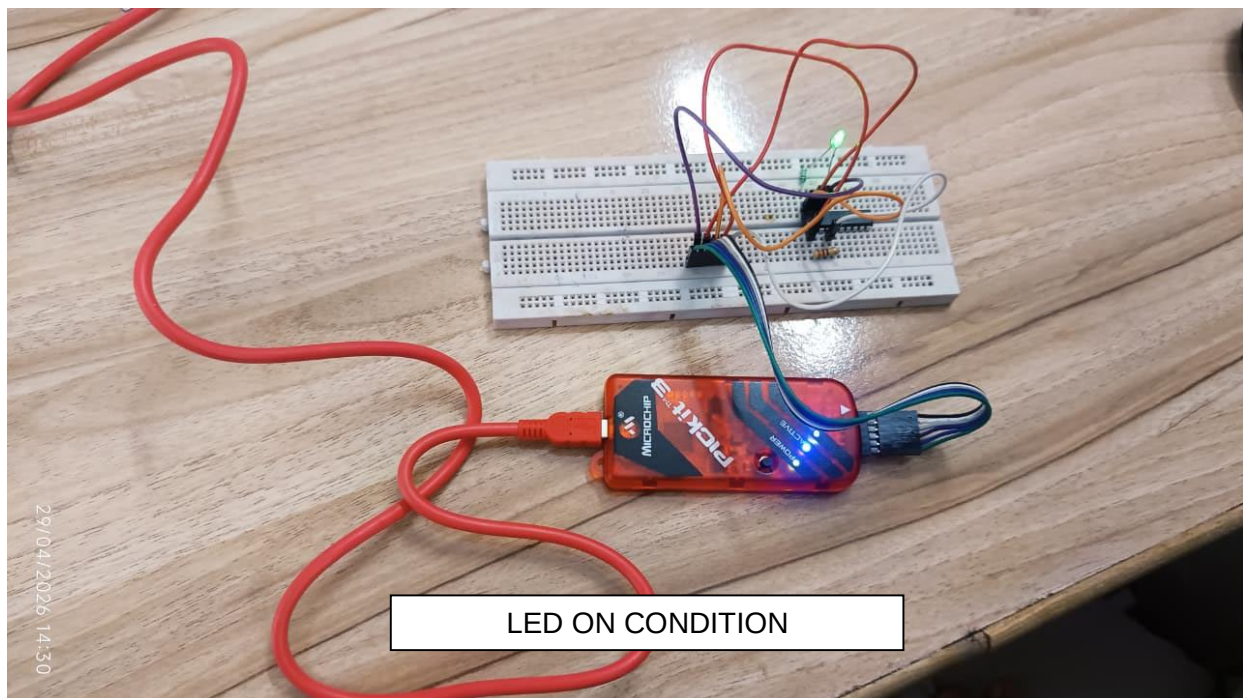
5. Procedural Steps

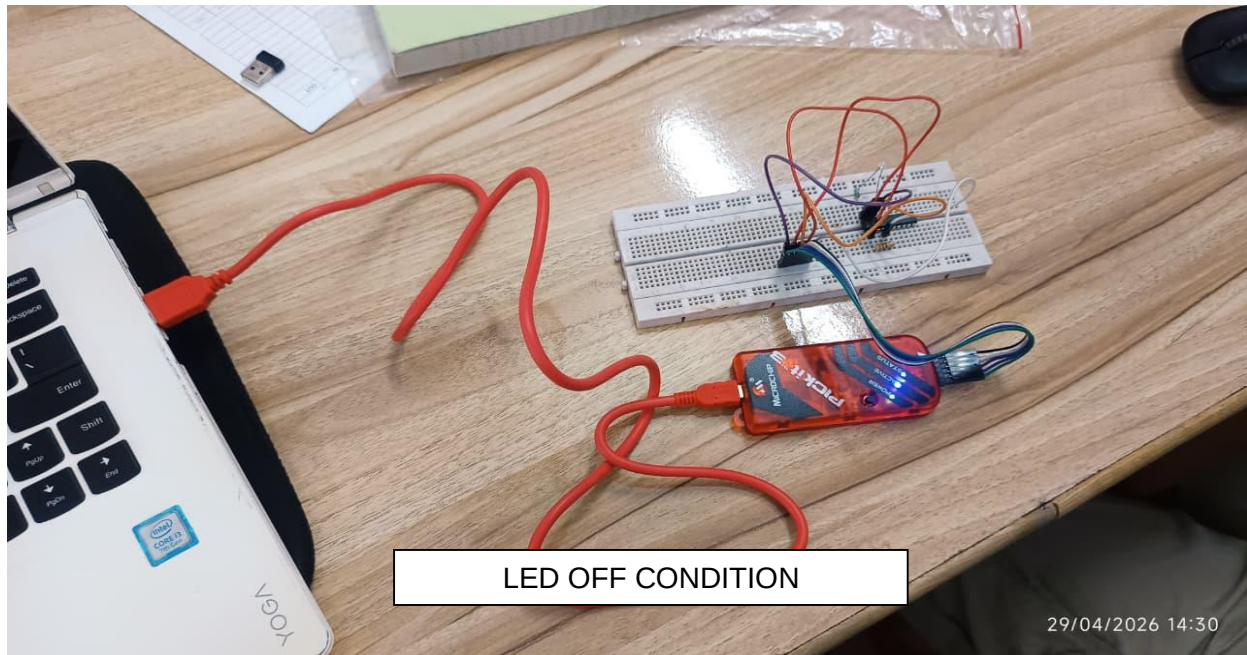
1. **Physical Assembly:** Insert the PIC16F690 into the breadboard and complete all wiring according to the Circuit diagram. Ensure the 10k Ω pull-up resistor is placed between Pin 4 and Pin 1.
2. **Project Initialization:** Open MPLAB X v6.20. Create a new Standalone Project, selecting the PIC16F690 as the device and PICKit 3 as the tool.
3. **Power Configuration:** In Project Properties > PICKit 3 > Power, enable "Power target circuit from PICKit 3" and set the voltage to 5.0V.
4. **Compiling:** Click the "Clean and Build" icon (Hammer and Broom). Verify "BUILD SUCCESSFUL" in the output window.
5. **Programming:** Click the "Make and Program Device" icon (Green Downward Arrow). The PICKit 3 will transfer the hex file to the microcontroller.

6. Hardware to Software Communication

The hardware supports only hex programming code, which makes the process somewhat cumbersome. Since we are using MPLAB X IDE v6.20, the source code must be compiled into a hex file before it can be flashed or executed on the hardware.

7. Prototype





8. Conclusion

Upon successful programming, the LED connected to Pin 16 will toggle at 1Hz (5 sec on, 5 sec off). This project confirms the proper setup of the development environment and successful hardware-to-software interfacing for the PIC16F690 platform.

9. Work Done By

Param Dohele

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